Segmentation of Bernini's Bronzes multiples using pairwise comparison of alignment

by

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Abstract

This thesis proposes an algorithmic approach to segmenting 3D scan data of bronze multiples, or sculptures comprised of the nearly identical segments connected together at slightly different angles. The result is intended to be useful for the Bernini's Bronzes technical study, a project focused on understanding and attributing the work of creating the many bronze multiples in Bernini's Oeuvre.

The proposed algorithm defines a cost function for segments based on the alignment error after aligning the segmented multiple to another base case multiple. Given that multiples are assumed to be made of the same components, a set of segmentations along the joints between the components should theoretically result in independent segments that can be perfectly aligned to their configuration in the base case multiple.

The results of running the algorithm on many sets of bronze multiples within and tangentially related to the Bernini's Bronzes dataset revealed promising potential and noticeable limitations. The resulting segmentation seams often aligned very well with know segment locations, and provided sensible results in multiples without known segments. However, it struggled to identify complex segmentation boundaries and deal with torus shaped geometries. More exploration into preprocessing steps, candidate segment exploration, and stopping criteria is needed.

Acknowledgements

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Contents

1. Introduction

The Technical Study of Bernini's Bronzes is a multi-disciplinary project investigating the many bronze sculptures attributed to Gianlorenzo Bernini. The study leverages expertise in the fields of art history, museum conservation, material science, and computer science, and aims to understand the contributions of the various anonymous foundry workers involved in the making of the bronzes, thus providing credit to people who were a part of art history and its development.

Much of this work relies on "multiples", sculptures wrought from the same original models created by Bernini. The indirect bronze casting method was often used during the Renaissance to create many nearly identical bronzes originating from a base model, and Bernini's bronze oeuvre contains many groups of multiples. The technical study assumes that any subtle differences between multiples, like the alignment of limbs or evidence of slightly warped or "worked" surfaces, can be attributed to the anonymous foundry workers, and as such establishing a system for comparison between multiples is of great interest.

The Bernini's Bronzes technical study has collected a diverse set of data on the available bronzes, including 3D light scans, high resolution photography, compositional analysis of both the bronze and the inner core material, and radiographic scans [1]. This multimodal data offers many angles at comparison between bronzes, and in the long term the project aims to combine and cluster the data using modern data analysis techniques. By leveraging many data sources, the study hopes to derive previously undiscovered insights into the history of process of the sculptures' creation.

The 3D light scans provide very useful data for comparison on their own, and are the focus of this paper. Macro differences between bronze multiples are easily captured by 3D data, and the resulting meshes can be transformed, aligned, and compared for volumetric and surface level deviations. Many key insights can be derived, including evidence of shrinkage from indirect bronze casting, identifying differing angles between segments of the multiples, and highlighting regions of likely hand-worked surfaces.

A challenging first step of analyzing the 3D data scans of bronze multiples is determining the difference in orientation of individual segments in these sculptures. Often the connections between distinct “segments” of the sculptures, for example the join between an arm and the torso of a human figure, contain different angles for each multiple. This difference is a barrier to many forms of comparison that require the sculptures to be aligned well, for example mapping deviation between two surfaces.

The source of these deviations can be traced back to the process of creating multiples. The final casting moulds formed from a base model often required piece-moulding, where distinct parts were joined together manually. Since this process is easily subject to alignment variance, and different multiples were often casted by different foundries, the orientation of these joins can differ between multiples. Thus, in addition to allowing for better alignment, identifying the locations of these joins is inherently useful for highlighting the process and input of the foundry workers.

Finding the locations where piece-moulded parts were joined can be conceptualized as a segmentation problem. Segmentation of 3D objects is a continually developing field with a wide array of applications, so research supporting this domain helps inform this task. Given this particular segmentation task has very specific criteria, a clear standardized methodology is not easily found, so there is room for innovation and adaptation to the task of analyzing Bernini’s works. Manual segmentation by knowledgeable experts is a solution that some studies on bronzes use, but there is potential for algorithmic approaches that may provide a different perspective or otherwise support the work of the experts.

The Bernini’s Bronze dataset in particular presents scans of many different multiples within a set. A simplifying assumption underpinning this research is that, for the most part, the individual segments of each multiple are identically casted from the original model created by Bernini, and that the most substantial difference between multiples is the orientation of the joins. Given this assumption, correctly locating and segmenting along the joins would minimize the alignment error between the meshes.

No existing segmentation approaches were found in research that operate on a dataset like this, where alignment between scans presents an optimizable error space. As such, a research gap exists, and this paper aims to address that research gap by developing segmentation methods for the 3D Bernini’s Bronzes dataset that leverage the presence of multiples. The underlying goal is to support the Bernini’s Bronzes Technical Study as a whole with data useful to their analysis, thus the key points are minimizing alignment error when comparing multiples and highlighting deviations. A clear objective is to reproduce, if not improve upon, the segmentation results achieved manually.

To achieve these points, a pairwise segmentation method is proposed, where candidate segmentation boundaries are evaluated by the reduction in alignment error between multiples after performing the cut. The pairwise comparison is unidirectional, where one multiple is arbitrarily assigned as the "base model" and the other is segmented and aligned to the base model. Alignment error is evaluated by aligning each segment separately, then evaluating the RMSE (Root Mean Squared Error) of the points across the segments when compared to the base model. Candidate segmentations are generated with semi-random planar cuts along the surface of the mesh.

Multiple measures are proposed for evaluating the success of a segmentation method in this space. Firstly, an optimal segmentation should minimize RMSE when aligning the resulting segments, and different segmentations can be compared by RMSE when they contain an equal number of cuts. Secondly, in contexts where the existing segments are known (if the segments were recorded during casting or were discovered via other analysis), the percentage of misclassified points is another measure to be minimized.

This research builds off of previous work done by Petra Alexson in developing methodology for analysis of Bernini's Bronzes scan data [2].

2. Literature Review

2.1 Lost-wax indirect casting and bronze multiples

During the Italian Renaissance, a predominant method of creating bronze sculptures was direct lost-wax casting, in which the initial shape and detail of the model was crafted using wax [3]. This wax model was covered in clay or plaster, creating a heat-resistant shell while leaving sprue holes for removing or adding material into the mould. The original wax model was then melted from within and drained out, to be replaced by liquid bronze. After removing the hardened bronze from the mould and removing artifacts of the process (like protrusions caused by filling from the sprue holes) the final product is achieved.

An innovation on this process is the art of indirect lost-wax casting, which allows for the original model to be reused to produce multiple sculptures [3]. For indirect casting, the original model can be made from any malleable material. In Bernini's case clay was used, though wax was another option. The model is coated with a plaster mould, and then the plaster mould is disassembled in parts, while taking care to not damage the original model. The resulting plaster segments are then filled with wax, individually recreating the parts of the original model. As the wax segments are extracted from the plaster moulds and put back together, this yields the wax intermodel, a near direct copy of the original model that can now be refined and subsequently used to create the bronze via direct casting. Naturally, plaster segments needed to be carefully planned such that they could be removed easily from the model while still accurately reflecting the shape, and tended to encompass meaningful sections of the piece like heads and arms.

Although indirect casting allows for multiple sculptures to be made from one original model, it introduces a few sources of disparity between multiples [3]. Immediately, the process of rejoining the wax segments offers many opportunities for deviation, as identically aligning and fastening segments at the exact original angles could not be expected. For sculptures of humans, this might manifest in the arms or head being tilted in slightly different directions between models. Fig. 2.1 visualizes the deviation of an arm between two models that are well aligned at the shoulder. It's clear to see that the blue multiple's forearm sticks out at a different direction compared to the red forearm.

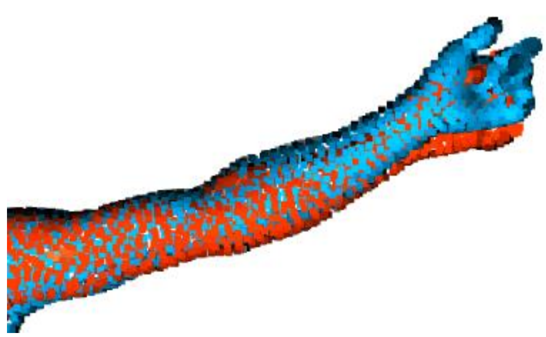


Figure 2.1: Deviating segments of multiples from the Bernini's Crucifixes set

Secondly, adjustments made to the wax intermodel would depend on the person working through the process [3]. Since fine details like drapery or hair must be simplified in the original model to allow for the plaster segments to be easily removed, the sculptor is often expected to deepen and refine details in the intermodel. Given multiples were often casted in different foundries by different skilled workers, this is a major source of subtle variation between sculptures casted from the same original model.

Additionally, entire segments may be absent from some multiples, either because the particular foundry decided not to cast them or the segment broke off at some point after the bronze was casted. As an example, in Bernini's oeuvre the Neptune multiples differ obviously, as one multiple holds a trident in its hands while the other lost its trident since casting.

Finally, the physical realities of the lost-wax process offer another subtle opportunity for deviation. Liquid substances, whether bronze or wax, tend to shrink as they cool, and amount of shrinkage is quantifiable and noticeable. Bronze alloys, depending on their composition, tend to shrink between 2-3% when cooling, while wax is extremely variable and depends on the type, temperature, and temperature difference between the wax and the mould [4]. The extent of the shrinkage can also depend on dimensionality, as long thin regions often shrink more proportionally along the cross section compared to the longer axis. Given indirect casting involves multiple stages of cooling with various substances, multiples tend to be noticeably smaller than the original model and can even differ in size from other multiples. In particular, if multiple intermodal stages are present, for example if lost wax casting is performed on a multiple, the shrinkage effect compounds yielding even smaller multiples.

2.2 Comparative analysis of sculpture 3D scans

The domain of segmentation for 3D scans of cultural heritage artifacts is not well represented in literature, particularly applying to the topic of sculpture multiples. However, 3D scan data has been used for comparison, alignment and deviation detection.

A technical study on Matisse's Aurora multiples leveraged volumetric comparison to determine the size differences between the sculptures [5]. Given that shrinkage occurs by predictable amounts when a cast is created from an existing model, this analysis was useful for determining the degrees of separation between each multiple and the base model, allowing for additional insights on the history of their creation. Volumetric comparison is a simple baseline method of analysis and is useful to keep in mind when designing alignment-based algorithms, as volumetric differences prevent perfect alignment, instead requiring the smaller surface to be perfectly centred within the larger surface.

Another study that employs 3D scan analysis focuses on Rosso's *Bambino ebreo* multiples [6]. Artifacts or seams found on many of the sculptures indicated that two moulds were used for casting, one for the face and one for the back of the head. After manually segmenting all of the multiples along this seam, alignment was performed on both the front and back of the 3D meshes, and various insights were detectable by examining the deviation maps after alignment. For example, small deviations in the face of the Folkwang Bronze when compared to the Cologne Bronze possibly indicated minor damage of the plaster between casts, which if true informs the casting order of the bronzes. Though the study does not perform automatic segmentation, the manual segmentation was paramount to extracting this deviation information, as the fronts and back were often aligned differently between multiples, making it hard to align just the faces. The study demonstrates the power of segmentation for useful comparative analysis, and informs potential use cases for the proposed segmentation methods in this paper.

The 2013 paper on portrait sculptures of Augustus aims to determine meaningful regions for classifying different stylistic categories [7]. The segmentation is performed on sculptures that are not multiples, but otherwise share similarities as they depict the same person. Their framework performs clustering which attempts to isolate regions that differ in the same way between the existing styles, eventually resulting in a segmentation that highlights the most useful segments for categorizing the sculptures. The paper focuses strongly on the middle forehead hair, as this feature is commonly used to distinguish statue types. The algorithm successfully clusters this region and shows clearly how the hair styling differs between sculptures. Of the studies found while exploring the topic, this paper most closely fills the research gap identified above, as it also identifies deviation between similar scanned meshes, and uses this data for segmentation.

2.3 Iterative closest point as alignment criteria

A well-defined evaluation metric for comparison between multiples is necessary to judge segmentation results. Given two similar point clouds that mainly differ by rotation and translation, a useful statistic would be to minimize the Root Mean Squared Error (RMSE), effectively the average distance between a point on one model and the closest point in the comparison model Perfectly aligned models would return an RMSE of 0, though the realities of working with bronze multiples mean that the sculptures will never be entirely identical. To compare notable deviations between similar models, they must first be aligned in 3D space, which entails minimizing the RMSE.

The standard algorithm to perform alignment of point clouds is Iterative Closest Point (ICP) [8]. This algorithm, when given two roughly aligned point clouds, computes and converges on an alignment by computing the nearest points in model B to model A, then computationally determining the optimal rotation and translation matrix. This transformation matrix is applied to all points in model B, performing one stage of alignment. Naturally, as the points have moved the associations between points must be recomputed. The process is repeated iteratively until a local minimum is found. If the initial rough alignment is reasonably accurate the algorithm is very likely to find the global minimum. Notably, ICP preserves the fundamental shape of the point cloud, as the rotational and translational operations are applied to all points.

Effectively, for each iteration the algorithm computes R, a 3x4 matrix, that minimizes the average Euclidian distance between x_i, a quaternion containing the 3D coordinates of a point in model B, and y_i, the closest 3D point in model A.

$$\min_R \sum_{i=1}^{N} ||Rx_i - y_i||^2$$

ICP alignment and RMSE are easily applicable to the domain of segmentation, as the theoretical goal of segmentation in this context is to better align individual parts of the multiples. Splitting a point cloud into two or more parts and then applying ICP allows the parts to move independently, which will necessarily decrease the alignment error by providing extra degrees of freedom. Given the RMSE of the point clouds can be weighted and averaged to determine the overall RMSE after segmentation, this provides a reasonable metric for the success of a segmentation approach.

2.4 Co-segmentation methods

Co-segmentation is the simultaneous segmentation of multiple objects in a consistent manner, and aims use the relationship between similar objects to classify distinct, shared parts of a set of objects [9]. In essence, it assumes that there are more complex insights to be gained about the fundamental structure of objects by looking at multiple within a class, a concept that applies very well to the Bernini's Bronzes dataset. Many supervised and unsupervised techniques for co-segmentation have been explored in literature, though the domain of Bernini's oeuvre does not supply great amounts of training data and is unlabeled. Supplementing the dataset would be a great challenge as this segmentation task requires nearly identical 3D meshes that only differ on orientation and are of similar complexity to the real bronze scans. As such, unsupervised techniques will be prioritized.

The most applicable group of strategies for unsupervised co-segmentation is alignment-based approaches, which focus solely on global alignment of the segments as they are, clustering and establishing correspondences based on the dimensions and geometries of the models [9]. These methods attempt to optimize the geometric similarity between aligned segments, meaning they depend on the parts of each model being mostly identical, an assumption that for the most part holds for bronze multiples.

A simple methodology that uses an alignment-based framework is showcased in Golovinskiy and Funkhouser [10], wherein the task is formulated as an optimal clustering problem. The meshes to be segmented are first aligned such that similar sections are near each other. Then, a graph is constructed where edges between neighbouring faces are scored by their likelihood of being segment boundaries using information on shape and concavity. Specifically, the paper proposes a cost of an edge to be the length multiplied by the dihedral angle θ raised to some power α, as in the equation below. Inter-mesh edges are created based on the closest points between the aligned meshes, linking them together. With all the meshes now connected in one graph, a greedy hierarchical algorithm is used to search for ideal segmentations that minimize the error function of each cut along the edges, resulting in segments that span all the meshes.

$$Cost = \min\left((\theta/\pi)^{\alpha} l, l\right)$$

Other complex methodologies are more robust to discrepancies in orientation and size, for example Huang *et al.* [11] which formulates a multi-stage optimization problem that individually over-segments the meshes and then searches for an optimal configuration that maximizes geometric similarity between co-segments, where both intra-segment and inter-segment similarity are prioritized.

3. Methods

After exploration and iteration of many methods, the algorithm for segmentation posed by this research is a pairwise alignment-based segmentation. Prospective segments are generated semi-randomly with planar seams and selected to minimize an objective function based around the alignment error between two multiples after segmentation. Each component of the segmentation process is described in detail below.

3.1 Preprocessing Steps

The initial inputs are the 3D meshes derived from scans of the many bronzes present in the Bernini's Bronzes dataset. Generally, the meshes are assumed to be closed surfaces comprised of triangles, where each triangle meets exactly one other triangle at an edge and all triangles are contained within the same connected face-edge graph. Issues were encountered while designing the segmentation algorithms due to these assumptions being violated in the real meshes, highlighting the importance of preprocessing the meshes such that they behave as idealized surfaces when segmenting.

The first step performed is a downsampling algorithm to reduce the number of vertices and faces in the mesh. This is achieved using quadric based edge collapse, adapted from Garland and Heckbert [12]. The algorithm selects pairs of connected vertices that minimize a cost function and replaces the two vertices with a single vertex, then reconnects the faces and removes redundancies. The cost function is defined with respect to the deviation from the original mesh that would occur after collapsing the edge, and is fairly complex to compute. This paper makes use of an implementation in Meshlab, an open-source software system for processing and editing meshes [13]. The effects of the downsampling process on St. Agnes #1 from the Bernini's Bronzes dataset are illustrated in Fig. 3.1, where the left is the original mesh and the right is downsampled by a factor of 10.

Figure 3.1: Downsampling performed on the St. Agnes #1 mesh

The main purpose of downsampling is to reduce the amount of data to be processed by the segmentation algorithms. The main algorithm has a time complexity of $O(n^2 \log(n))$ for number of points n, so reducing the topological complexity of the mesh presents major speedup potential. All meshes are downsampled by a factor of 10, resulting in 20,000-30,000 vertices in most cases. It is assumed that the fine detail removed by downsampling does not after the alignment process, as the general locations of points in the point clouds do not change, and still reflect the alignment deviations in the base models.

After downsampling, all meshes are adjusted to ensure they are well-formed closed surfaces. Though all scans in the Bernini's Bronzes dataset are intended to accurately capture the real-life objects, artifacts of the scanning process are clearly present in most models. Holes in the meshes represent regions that are missing scanning data, likely due to the selected angles of capture missing specific surfaces. Faces and vertices completely unconnected from the main graph of faces are also present in some meshes, likely an artifact of the alignment process between different angles of capture. Finally, multiple non-manifold edges are present, defined as edges where 3 or more faces meet.

Since all of these irregularities exist in the Bernini's Bronzes meshes, all are manually removed. Any disconnected faces are isolated and deleted, non-manifold edges are detected and fully removed, and a hole closing algorithm in Meshlab is used to connect the vertices surrounding a hole with faces. Fig 3.2. shows disconnected faces resulting from poor scan alignment of the Horse #3 mesh, while Fig. 3.3 visualizes hole filling in the St. Agnes #1 scan data.

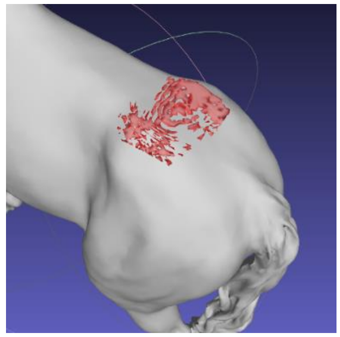


Figure 3.2: Unconnected mesh artifact from Horse #3 scan

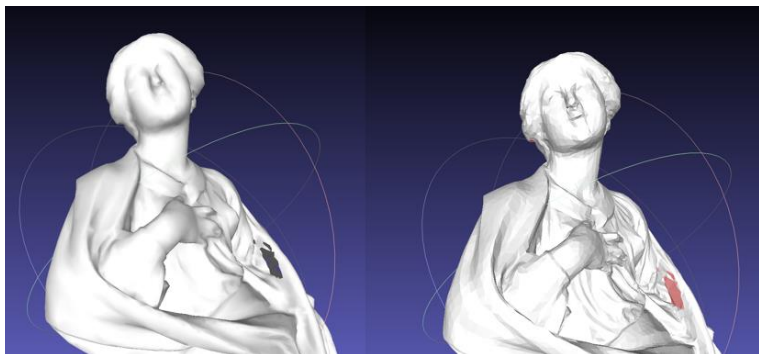


Figure 3.3: Filled hole, pictured in red, on the St. Agnes #1 mesh

Finally, given a set of 2 or more multiples to be analyzed, all meshes are roughly aligned spatially. The ICP algorithm requires that point clouds are nearly aligned to ensure that its initialization is near the global minimum, otherwise it's less likely to converge to a sensible alignment.

3.2 Segmentation Candidate Generation

After preprocessing, the next stage is to generate a set of potential segmentation seams on the surface of the meshes. For a pairwise comparison, one of the two multiples is defined as the **base model** used strictly for comparison, while the other is the **test model** for performing segmentations. Many theoretical regimes for defining potential segmentations are available and could be slotted into the algorithm in this stage, thus this step offers the most potential for iteration.

The method of candidate generation used for this research is fairly simple and relies on sampling. Each point in the test model is individually used to define a segmentation seam, meaning that the number of potential cuts on the model will be equal to the number of points.

To define a segmentation seam, a plane is defined that contains a vertex on the mesh. The normal vector to the mesh surface at that point is set as one of the unit vectors of the segmentation plane, while the other unit vector is randomly generated. This method ensures that the segmentation plane intersects directly down into the mesh, though given the second unit vector is random there is no assurance that the segmentation seam is semantically meaningful. Visualized in Fig. 3.4 is a segmentation plane intersecting the surface of a model, originating from a red point.

Figure 3.4: Visualization of a segmentation plane on a point cloud

Given the random aspect of the sampled planes, most segmentation planes generated will not be semantically meaningful. As an example, typically the most meaningful segmentation of a cylindrical object would be perpendicular to its long axis, producing a circular cross-section. In Fig. 3.5, many possible segmentations using the pictured point and unit vector are present, but only the green seam cuts approximately perpendicular to the long axis. A fundamental assumption of this methodology is that, given enough sampled segmentation seams, it is probabilistically likely that meaningful segmentations will be found. Hence, one seam is generated for every point, in meshes with tens of thousands of points.

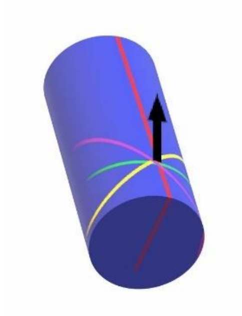


Figure 3.5: Visualization of many possible segmentation seams on a cylinder

To increase the probability of finding the best segments even more, points could be sampled multiple times at the cost of increased runtime. As is, the amount of potential candidates used did not seem to be a major limiting factor in the results of the algorithm.

Once a segmentation plane is selected, the seam is determined by tracing the intersection of the plane with the surface of the mesh until to forms a closed loop. Tracing begins at the point directly intersected by the plane. A segmentation plane may intersect the mesh at multiple regions, but only the intersection containing the original point is selected. Notably, if the segmentation seam intersects a hole in the mesh, a closed loop on the surface can not be formed, hence why the preprocessing step fills all holes.

3.3 Segment Cost Function

After generating the set of segmentation seams, the main runtime of the algorithm is devoted to evaluating the effectiveness of each segmentation and choosing the best set. The cost function of a particular segmentation is defined as follows. Given the two resulting segments X_1 and X_2 after cutting along the seam in the test model X, let R_1 be the rotational/translational matrix returned by ICP aligning the points of X_1 to the base model Y, and let R_2 be a matrix for aligning X_2 to Y. If f(X,Y) computes the sum of the squared distances between points in X and the closest corresponding points in Y, the cost is defined as:

$$Cost = f(R_1X_1, Y) + f(R_2X_2, Y) - f(X, Y)$$

Effectively, this measures the reduction in squared error after performing the segmentation and allowing the individual segments to be align to the base model. Since these quantities are proportional to the number of points in X, the cost function prioritizes segmentations that reduce alignment error in larger groups of points. f(X,Y) will always be larger than the other two terms, as adding a degree of freedom by splitting the segment necessarily improves the ability for ICP to align the point clouds.

Pictured in Fig. 3.6 is a visualization of all seams generated and the costs of these seams on Crucifix #1 when compared to the Crucifix #2, both multiples within the Bernini's Bronzes dataset. Green seams represent the smallest costs, while red seams represent the largest costs. From the figure, it's clear that the best single cut that can be made to reduce the alignment error lies somewhere around the knees of crucifix, indicating that the lower legs deviate greatly between the multiples.

Figure 3.6: Cost of various segmentation seams on Crucifix #1

This cost function can be adapted to prioritize additional criteria. For example, Golovinskiy and Funkhouser uses a measure of concavity as a part of their segmentation cost function, assigning low costs to seams that lie on highly concave edges (where concavity is the dihedral angle between the faces that comprise the edge). This reflects the physical reality of segments in objects, as meaningful segments often meet at concave angles. Adapting this principle to the methods of the paper above, a concavity score can be computed for each part of the segmentation seam on a particular face of the mesh.

To compute the concavity score for one segment of the seam, the similarity of each edge in the face is computed with the normal vector of the plane. If similarity is high, the edge does not align with the segmentation plane (and thus the seam) so its weight is lowered. The full equation used is provided below, where n is the normal vector of the plane, e is an edge of the face, and C_e is the concavity of that edge.

$$Concavity\ of\ Seam\ Segment = \frac{\sum_e((1 - |e \cdot n|)^2 \cdot C_e)}{\sum_e(1 - |e \cdot n|)^2}$$

If C is defined as the weighted average concavity of every portion of the segmentation seam, weighed by the length of each segment, an adapted cost function is defined as follows. Note that C varies from 0 to 1, where 0 is most concave and 1 is least concave, thus more concave C values increase the value of f(X, Y), reducing the cost. The value is C is raised to the ½ to reduce its overall effect on the cost.

$$Cost_C = f(R_1X_1, Y) + f(R_2X_2, Y) - \frac{f(X,Y)}{C^{1/2}}$$

3.4 Repeated Segmentation

The cost function above can be used to split the test model into two segments by selecting the seam with the lowest cost. Isolating either of the two resulting segments, the cost function can be recomputed on just that segment, testing seams on it and computing the further reduction in alignment error after splitting on those seams. Given that many bronzes in Bernini's oeuvre contain more than two segments, a strategy is defined to further segment the mesh. A sample of the process performed on a set of multiples is provided in Fig. 3.7.

1. Compute the cost function for every potential segmentation seam on the test model. If a segment seam crosses an existing cut (spans multiple existing segments) it is disregarded, each segmentation must split exactly one segment into two.
2. Select the seam with the minimum cost and perform the split on that segment of the test model. The number of segments has grown by one.
3. Repeat steps 1 and 2 until some stopping criteria is reached

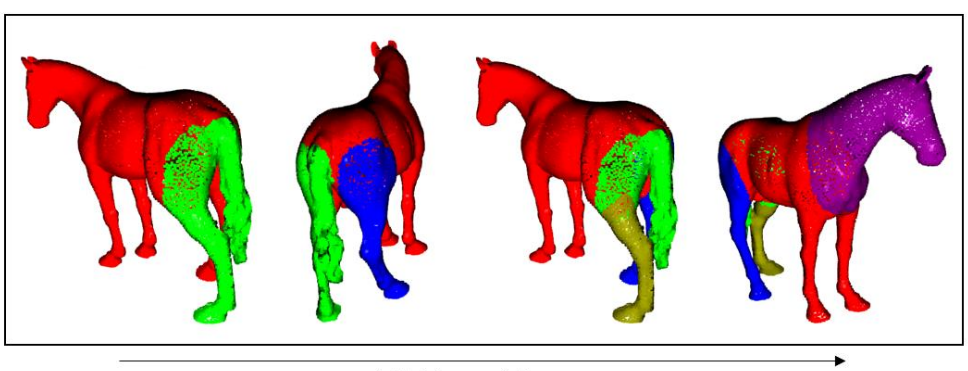
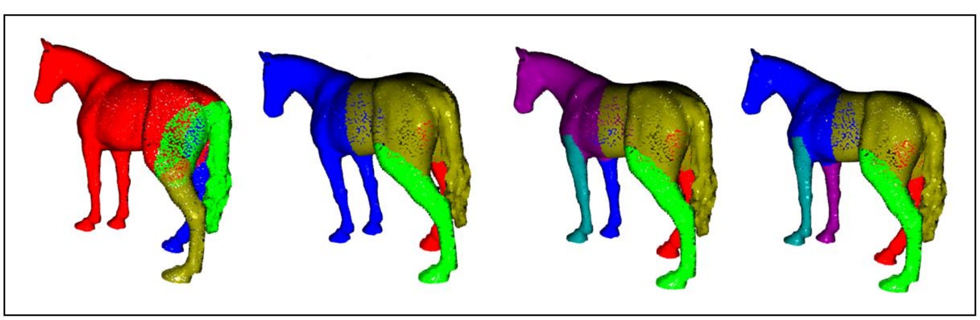


Figure 3.7: Visualization of segmentation adding procedure on

This strategy employs greedy choices, always selecting the best new segmentation to add given the current segments. However, the problem space does not satisfy the greedy choice property, as the best set of two segmentation seams may not include the best singular segmentation seam. In Fig. 3.7, the first segment is defined to include both the tail and a hind leg despite the components being separately casted, likely because the deviations of both parts are slightly correlated (applying the same transformation matrix to both reduces the error in both). This first segmentation limits possible subsequent segmentations as they must not cross this first seam. It is particularly hard to find a seam that separates the tail from the upper hind leg without crossing the green-red seam.

As such, an addition was made to the algorithm above. After every two new segments are added, a re-evaluation pass is conducted, where one by one each current segmentation seam is refined. Specifically, the two segments bordering the seam are merged (completely removing the segmentation) and the cost function is recomputed globally. A new segmentation seam that minimizes the cost function is then added to replace the original one. If the original was a good choice given the current configuration of segments, it will likely be reselected. However, this process allows for bad initial segmentation choices to be replaced.

In Fig. 3.8, the same mesh is split while utilizing re-evaluation passes. The results of every two new segmentations are pictured, both before and after the re-evaluation step is run. Notably, after the re-evaluation pass the initial poor segmentation seam involving the hind leg and the tail is replaced, better reflecting the realities of the model. Overall, including the re-evaluation step improved performance greatly.



Four Segments, Before Refinement | Four Segments, After Refinement | Six Segments, Before Refinement | Six Segments, After Refinement

Figure 3.8: Segmentation progression including re-evaluation steps

3.4 Measures of Performance

Multiple metrics are proposed to evaluate the effectiveness of a segmentation result on a set of multiples. These metrics prioritize accuracy where true segmentation results are known, and otherwise focus on reducing alignment error.

Within the Bernini's Bronzes dataset and related test datasets, there are a few multiples for which known segment locations exist. The Horse set is a group of bronze multiples casted in modern times to serve as test data for much of the analysis in the technical study, thus the approximate locations of every segment joint are known, examples of which are pictured in Fig. 3.9. On the other hand, Bernini's Matilda sculptures were casted long ago, but modern analysis from art historians and conservators reveal likely mould lines on specific Matilda multiples as pictured in Fig. 3.9, which can be tentatively treated as true segments for the purposes of testing the algorithm.

Figure 3.9: Left: Horse segment moulds   Right: Drawn mould lines on Matilda sculpture

Where known data exists, the simplest way to evaluate a segmentation is to compare the percentage of correctly classified points. To do this, a set of one-to-one associations between the true segments and the algorithm's segments are drawn. The number of associations will be the smaller of the total segments in the true segmentation and the total in the algorithm's segmentation. The best possible set of associations between segments is used to evaluate success, shown in Fig. 3.10 for Horse #2 mesh. To find the best possible set of associations every combination is checked, though given the factorial time complexity this strategy does not scale for large numbers of segments.

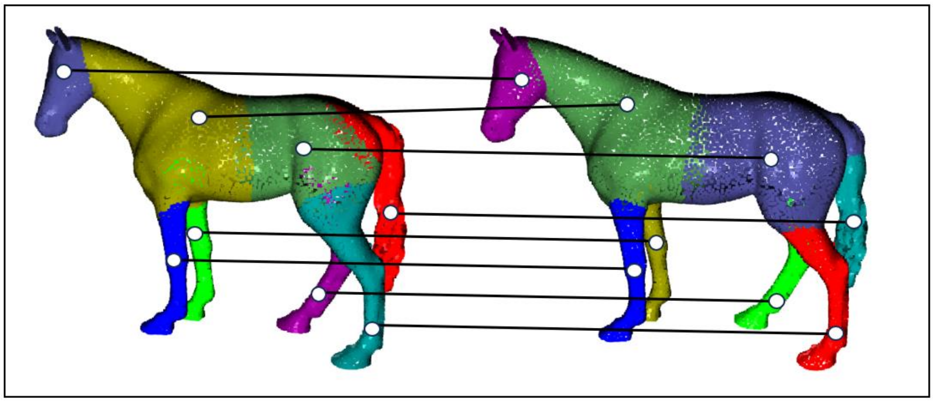


Figure 3.10: Optimal set of associations between two segmentation schemes

When no true segmentations are available, alignment error is used as the main evaluation tool. The RMSE of all points in the test model compared to the base model after segmentation and alignment should indicate how well segment joins are detected, as rotating parts of the test model exactly at these joins should align the identical components, maximally reducing the RMSE. Notably, the magnitude of the RMSE itself is not a meaningful measure as it depends on the fundamental geometries of the multiples, for example their size, their size discrepancy due to shrinkage, and hand-worked regions causing further deviation. Instead, RMSE should be used to compare different segmentations. Note that only segmentation sets with equal numbers of sections can be compared, as the optimal RMSE monotonically decreases with the number of segments (additional degrees of freedom necessarily allow for closer alignment).

Though more exploration is required, RMSE can also serve as a reasonable stopping criterion for the segmentation algorithm. For multiples with unknown segments, choosing when to stop the segmenting process is a challenge, and defining numerically the point of diminishing returns can be achieved by tracking the RMSE with each new segment, with the goal of recognizing when further segmentation isn't worth it. Fig. 3.11 illustrates the diminishing returns on MSE (Mean Squared Error) reduction when performing segmentation on the Horse #2 model. Note that this paper does not propose an exact methodology for stopping criteria due to the complexity of defining a rule that works in most contexts with the limited data available.

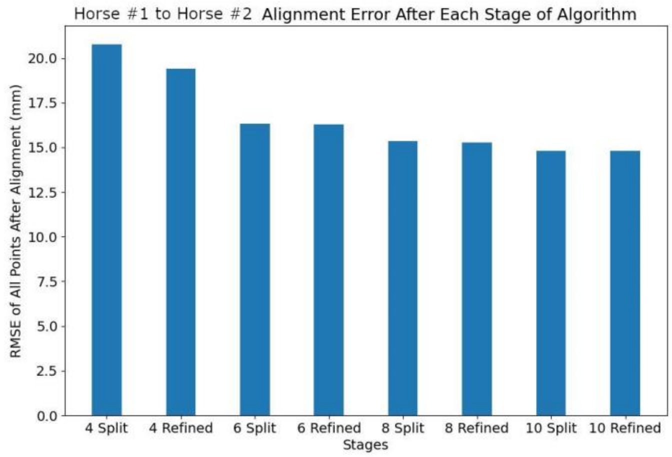


Figure 3.11: MSE of Horse #2 at each stage of the segmentation algorithm

4. Results and Discussion

In the following section, the methods outlined above are performed on four sets of multiples selected from the Bernini's Bronzes dataset and other test datasets created for the technical study. The four sets of multiples are the Horse set from the test datasets, and the Matilda of Tuscany, St. Agnes, and Crucifix sets from Bernini's oeuvre. Prioritization on which sets to study was determined by quality of the data scans and the expected number of segments, as well-formed scans with a lot of potential wax-to-wax or other jons are most useful to the project.

For the Matildas and the Horses, known segmentations are present, so analysis for them will include correct classification % as a metric. All sets will be evaluated for MSE (RMSE squared) as the segmentation is performed. By default, segmentation is performed without using the concavity measure in the cost function, though some examples with it are included. In many cases both directions of a comparison are provided (A to B and B to A). Further analysis on the challenges present and other uncertainty is interspersed throughout.

4.1 Horse Multiples

The Horse multiples are a set of bronze casted sculptures created in 2022 in support of the Bernini's Bronzes technical study. It consists of four multiples: Horse #1, Horse #2, Horse #3, and Horse #4, all pictured in Fig 4.1. The original model was casted in pieces, and when putting together the segments they were intentionally placed at slightly different angles for each of the multiples. Given that the locations of the seams between segments are explicitly known, and the segments deviate in similar ways to the multiples from Bernini, this dataset provides a great proxy for evaluating the performance of the algorithms. Fig. 4.2 shows an approximation of the true segment boundaries on Horse #2.

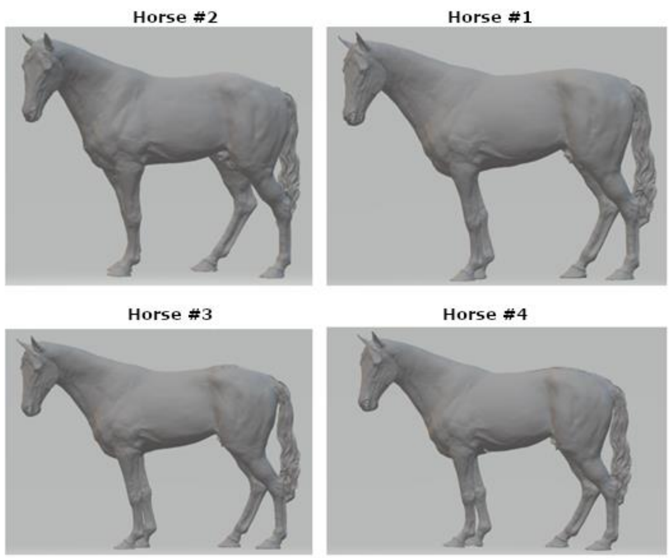


Figure 4.1: Horse multiples

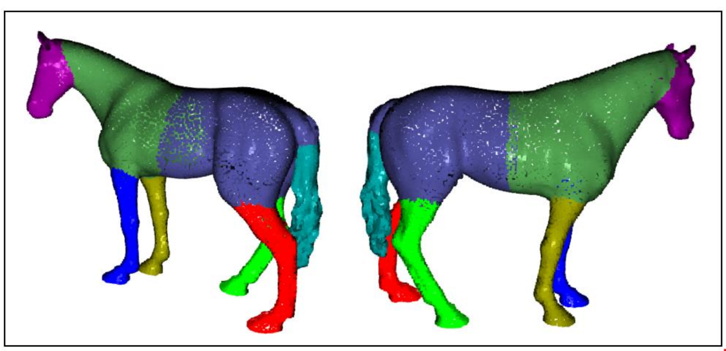


Figure 4.2: Approximate true segmentation of Horse #2

4.1.1 Segmentations

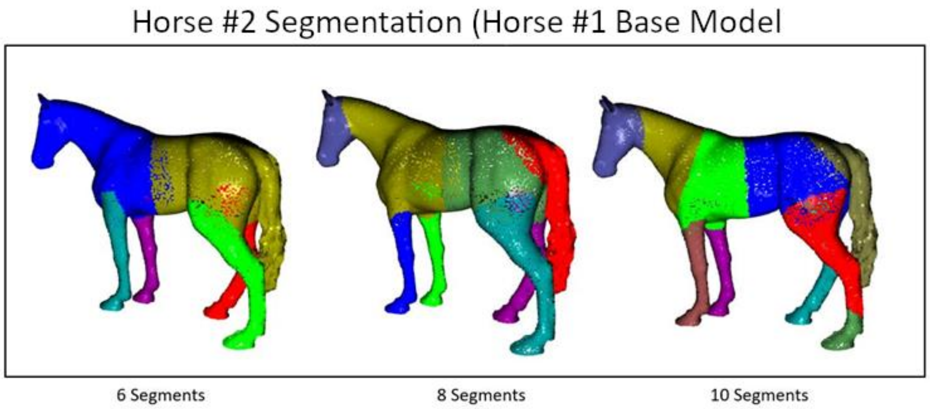


Figure 4.3: Horse #2 to Horse #1 segmentations

Focusing on the Horse #2 segmentation in Fig. 4.3, the 8-segment result very closely resembles the true segment boundaries. All semantically meaningful sections are correctly classified, and only the exact boundaries differ. The most accurate seams are drawn on the head and front legs, while the back legs and tail are less accurate as they include much of the rump of the horse.

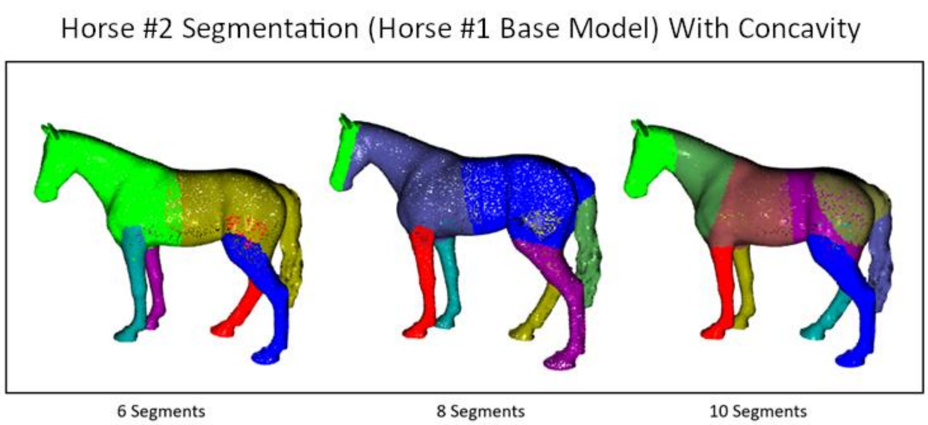


Figure 4.4: Horse #2 to Horse #1 segmentations factoring in concavity

With the concavity augmented segmentation in Fig. 4.4, the tail and back legs are considerably more accurate, in large part because the correct seams lie on regions of high concavity (e.g. the narrowing region of the tail). Including the concavity data biases the segmentation towards these more meaningful regions. The segmentation on the head for the 8-

segment model is noticeably worse on the other hand, since the seam passes through the highly concave mouth of the horse. Given that the 10-segment result immediately fixes this issue, it's likely the two seams are very close in cost, and reducing the effect of the concavity in the cost function would fix the issue in the 8-segment version.

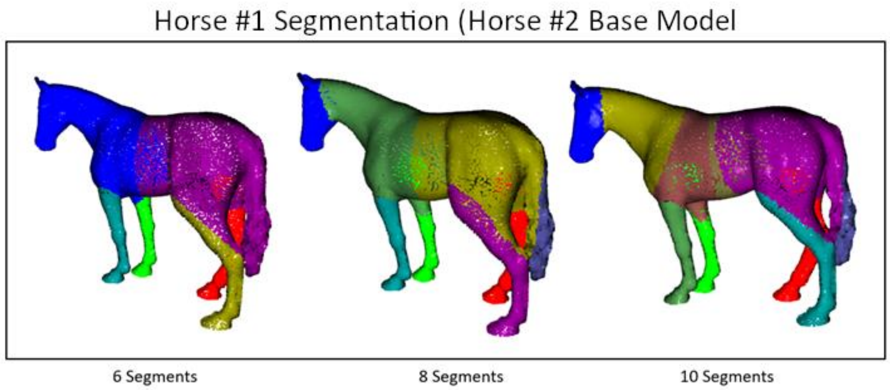


Figure 4.5: Horse #1 to Horse #2 segmentations

Comparing either result to the Horse #1 segmentation in Fig. 4.5 shows that this process is notably not symmetrical. Segmenting model A based on alignment to model B does not produce the same segmentations as segmenting model B based on alignment to model A. In particular, Horse #1 aligned to Horse #2 segmentation shows a very unintuitive set of segmentations around the tail and back leg.

The reason for this artifact is that, in the Horse #1 mesh, the back left leg is attached to the tail by a thin junction. This topologically produces a torus shape, and torus shapes are a fundamental weakness of the algorithm. When a potential seam to be evaluated cuts the torus shape, say at the upper leg or near the base of the tail, this cut does not split the model into two components, as every point in the model is still connected going the other way around the loop of the torus. Since the model is not split, there is only one segment, and no improvement in alignment can be made. All seams splitting a torus shape will fail to reduce RMSE, and will never be selected by the cost function.

Despite this flaw, the segmentation on the bronze is still interesting to analyze. The selected segments in this problem area very clearly contort around this junction point, with the two segments including as much of the tail and as much of the leg as possible without cutting the torus. To some extent, this illustrates the robustness of the candidate selection phase, as these planar seams seem almost intentionally selected to handle this problematic region.

Included below are a couple additional samples of segmentations using the Horse #3 and Horse #4 multiples. The results of the Horse #3 to Horse #1 segmentation in Fig. 4.6 are very similar to Horse #2 to Horse #1, while the Horse #4 segmentation in Fig. 4.7 is much less accurate. Part of the issue with Horse #4 is the size difference, as it is noticeably bigger than Horse #1due to shrinkage differences. This causes the algorithm to select large regions of the torso as sections and move them inwards to align with the smaller Horse #1 base model, adding noise to the process. The preprocessing step does not normalize the multiples with respect to size, though given that the size difference clearly hurts accuracy in this case there is a reason to consider it.

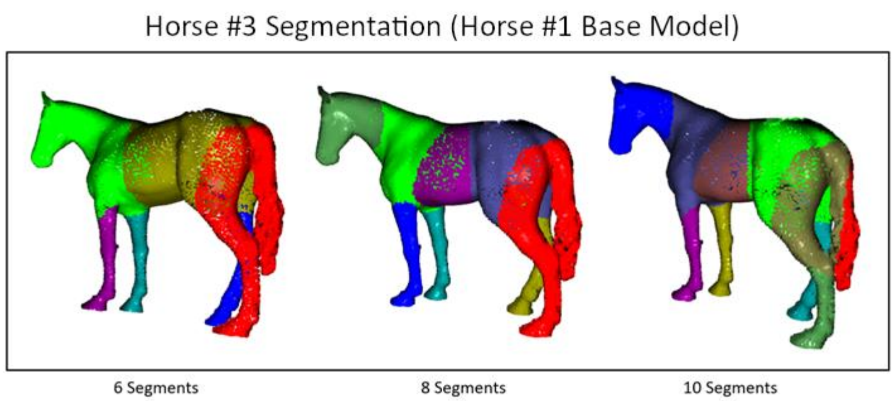


Figure 4.6: Horse #3 to Horse #1 segmentations

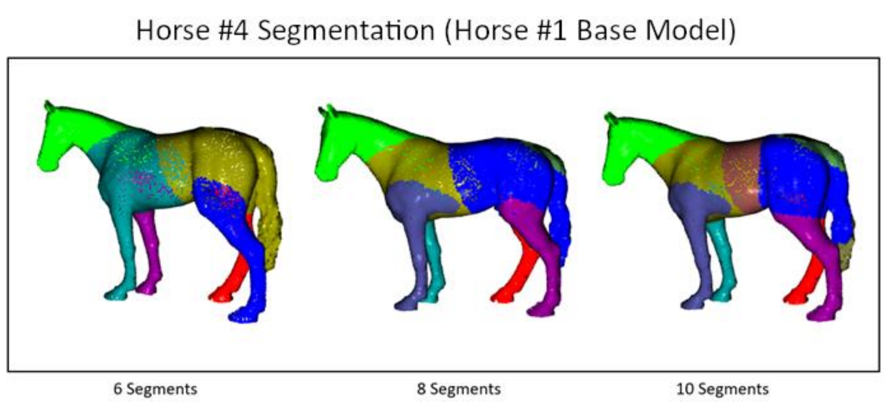


Figure 4.7: Horse #4 to Horse #1 segmentations

4.1.2 Evaluations

These multiples contain known segmentations, so the segmentation accuracy of each pair of multiples can be evaluated for each possible number of segments. The following table illustrates the values for each.

	4 Segments	6 Segments	8 Segments	10 Segments
Horse #2 (Horse #1 Base)	60.16%	69.40%	84.50%	71.48%
Horse #2 (Horse #1 Base) Concavity	61.22%	68.50%	90.94%	73.68%
Horse #2 (Horse #4 Base)	55.15%	56.43%	62.79%	59.36%
Horse #1 (Horse #2 Base)	62.24%	72.19%	87.46%	73.40%
Horse #3 (Horse #1 Base)	61.44%	63.09%	60.87%	65.46%

Firstly, given that there are 8 real segments it's natural that almost every pair of multiples returns the best results when 8 segments are used. In one case, the original wax segmentation, this does not hold, likely because the selection of cuts is poor for those 7^{th} and 8^{th} segments. Like Horse #3, Horse #4 also shows major size differences, so the lack of accuracy is fairly explainable.

Only comparing segmentations that don't use concavity, interestingly the Horse #1 segmentation with a Horse #2 base performs the best for each number of segments, despite the issue with the leg-tail junction. This is because the Horse #2 segmentation without concavity has the tail segment which includes much of the rump of the horse inaccurately, while the Horse #1 segmentation only performs poorly near the junction.

Of course, using the concavity measure the Horse #2 with Horse #1 base segmentation performs very well, showing the highest accuracy for 8 segments. In this case using concavity is shown to improve accuracy, though more concavity data is needed for some of the other pairs.

MSE for each stage of the segmentation process were recorded for many of the pairs of multiples, some of which are visualized in Fig. 4.8. For terminology, each stage is denoted by a number representing the number of current segments, "Split" notes that the value was recorded before performing the re-evaluation step, while “Refined” notes that value was recorded after re-evaluating all the current seams.

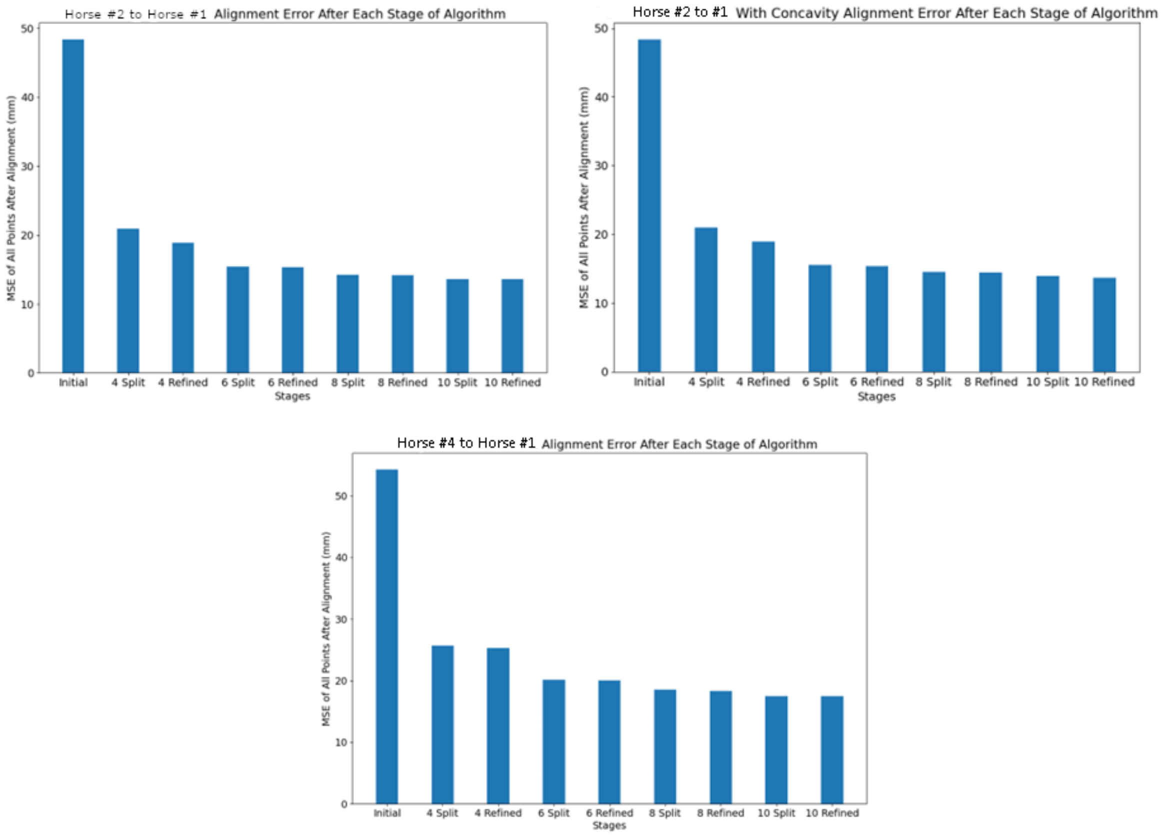


Figure 4.8: MSE at each stage of segmentation for various Horse multiple pairs

Of the three pictured, the baseline MSE before segmentation is similar for Horse #2 to Horse #1 compared to Horse #4 to Horse #1, mainly because the magnitude of the MSE is dominated by heavily misaligned components for both. However, as segmentations are made the Horse #4 to Horse #1 MSE does not reduce by nearly as much, mainly because the size difference between Horse #4 and Horse #1 prevents a good alignment.

Comparing the Horse #2 to Horse #1 segmentation with and without concavity, the MSE is very slightly worse for the larger number of segments. This is because the loss function, in prioritizing high concavity seams, slightly reduces the importance of minimizing MSE. As seen in the accuracy table, this trade-off improves the actual segmentation at the expense of worse MSE.

Another interesting element to analyze is the effect re-evaluation has on MSE. For smaller segment counts, specifically 4 segments, re-evaluation yields a reduction in MSE, indicating that the first couple segmentations are often suboptimal for the Horse models. This may be a feature of this specific set of multiples or a general trend. For larger segment numbers, MSE shows no clear trend in rising or falling, indicating that the later re-evaluation phases have much less utility.

Overall, the prospect of finding stopping criteria with this data seems promising but inconclusive. It's clear that there are diminishing returns for each new segment added, but numerically defining a threshold at which point segmentation should stop is a challenge given that both the MSE and the $MSE/MSE_{Original}$ approach different values.

4.2 Matilda of Tuscany Multiples

The Matilda of Tuscany multiples belong to the Bernini's Bronzes dataset, and are the set with the largest number of multiples. There are 7 bronzes with scan data in the dataset all originating from the same mould created by Bernini. For the purposes of this paper, 3 of the 7 have been used for segmentation so far: Matilda #1, Matilda #2, and Matilda #3, pictured in Fig. 4.9. Using identified mould lines on the surface of a Matilda and other knowledge from the team, "true" segmentation are defined used for accuracy calculations. The true segments on Matilda #3 are pictured in Fig. 4.10.

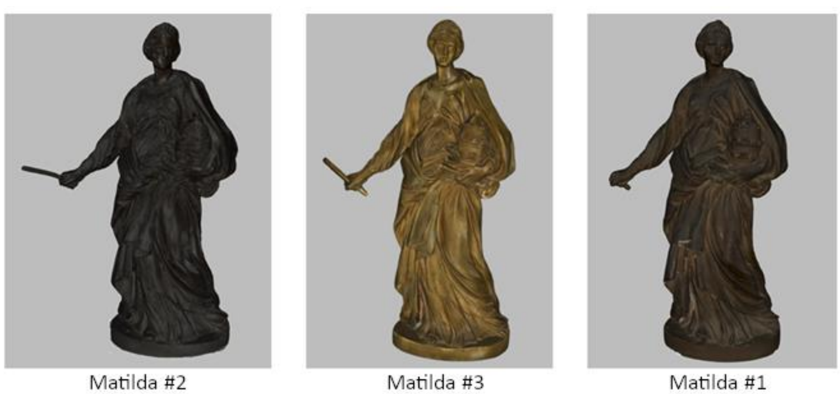


Figure 4.9: Matilda of Tuscany selection

Figure 4.10: Approximate true segmentation of Matilda #3

4.2.1 Segmentations

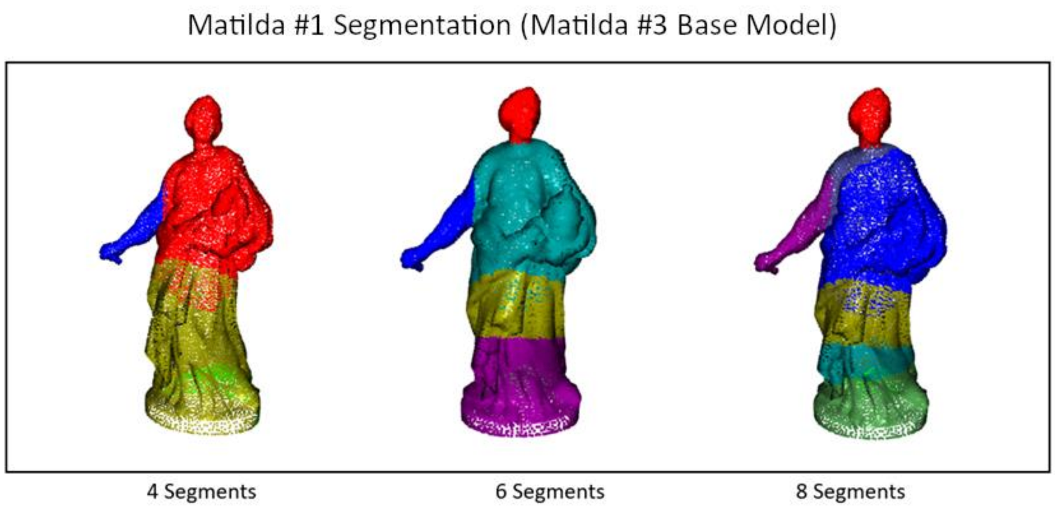


Figure 4.11: Matilda #1 to Matilda #3 segmentations

The Matilda #1 segmentation using Matilda #3 as a base model provides the closest approximation to the estimated true segments, though it requires 6 segments to properly find these seams. At the 4-segment stage, the arm is segmented separately unlike in the true segments, and the back of the base (visible behind the yellow point cloud) is also its own segment. For the base, during the preprocessing stage some manipulation was required to remove poor quality scan data like the floor around the model and the inner cavity, so any discrepancies down there may partially be attributed to that work. For the arm, it's possible that there is a true segment there, it was harder to confirm with sources from the technical study so it was not included but there is strong evidence that the arm angles are different. Regardless, at the 6-segment phase both seams along the dress are successfully approximated, as is the head.

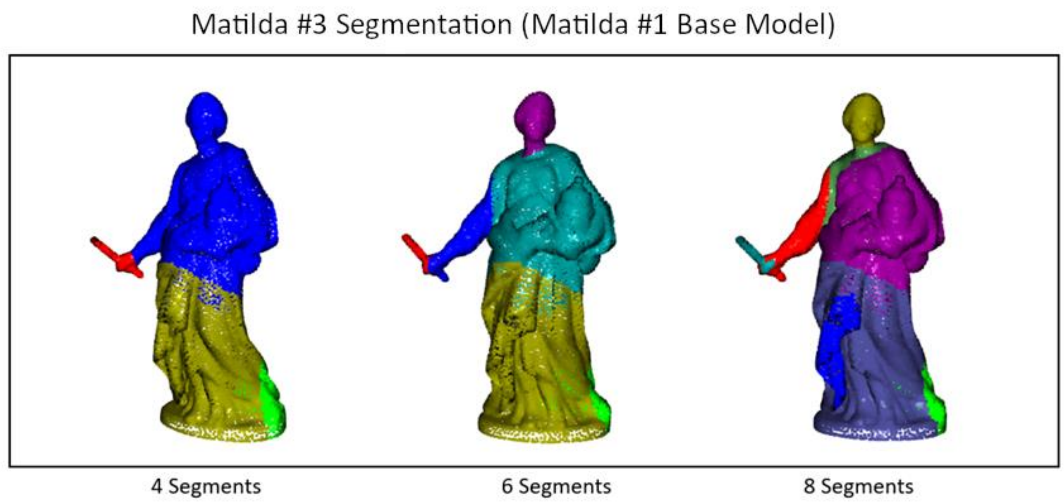


Figure 4.12: Matilda #3 to Matilda #1segmentations

Looking at the reverse direction in Fig. 4.12, the segmentation is generally less successful but also identifies some key points of interest.

Firstly, a key asymmetry exists, as Matilda #3 holds a baton while Matilda #1 does not. When segmenting Matilda #3, the baton represents a major source of alignment error and is immediately segmented and translated to align with the arm. In any case where components are missing in one of the two multiples, directionality will matter greatly, as the mesh without a component does not have a component to realign, and thus will not be segmented at that location.

Because of the baton, and the base of the model once again containing an additional segment, this segmentation does not identify the second real seam on the dress, though it otherwise identifies the other key seams in the true segmentation. The arm is again treated as its own section, once again indicating that the angles are different between the multiples.

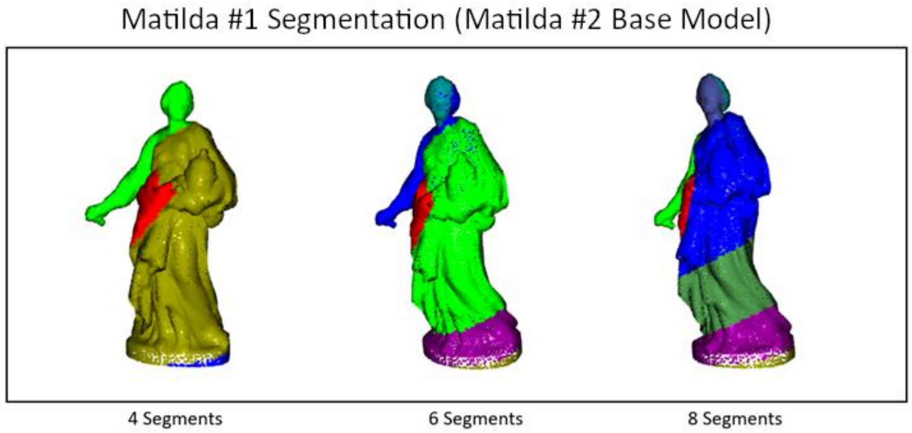


Figure 4.13: Matilda #1 to Matilda #2 segmentations

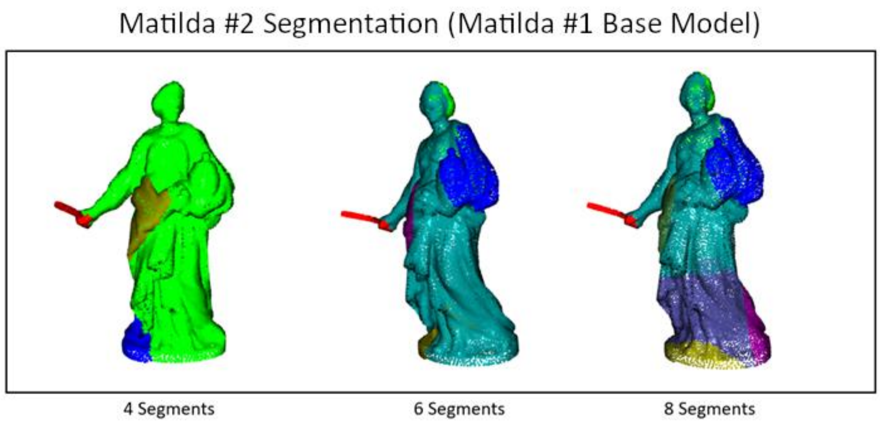


Figure 4.14: Matilda #2 to Matilda #1 segmentations

Both comparisons between Matilda #1 and Matilda #2, present in Fig. 4.13 and Fig. 4.14, show less effectiveness at identifying relevant seams. At the 8-segment stage the Matilda #2 segmentation identifies one of the seams, and Matilda #1 gets close, but they both prioritize other regions around the base and at other folds in the dress. Interestingly, some more complex mould lines identified on the Matilda sculptures somewhat align with these segments, so though they were ignored when creating the "true" segmentation these seams may be meaningful. Also, these comparisons show similar trends to the Matilda #1-Matilda #3 comparisons with the baton asymmetry and the base being a region of deviation.

4.2.2 Evaluations

Noting that there is much more uncertainty that the “true” segmentations are correct and complete, the segmentation accuracies for each pair of multiples is presented below.

	4 Segments	6 Segments	8 Segments
Matilda #3 (Matilda #1 Base)	67.21%	67.36%	59.15%
Matilda #1 (Matilda #3 Base)	64.64%	81.82%	64.59%
Matilda #1 (Matilda #2 Base)	40.84%	48.47%	55.34%

Confirming visual intuition, the 6-segment output for Matilda #1 compared to Matilda #3 produces the best accuracy, with the only major misclassified regions being the arm and part of the base. The Matilda #1-Matilda #2 performance in recreating the “true” segments is the worst of the three, though it’s clear that there may be more interesting analysis on the true locations of segments in that comparison. Unlike the Horses, all 3 comparisons perform better when oversegmenting (more than the 4 “true” segments), though this may indicate that the true segments being used are inadequate for describing the process of creating the multiples.

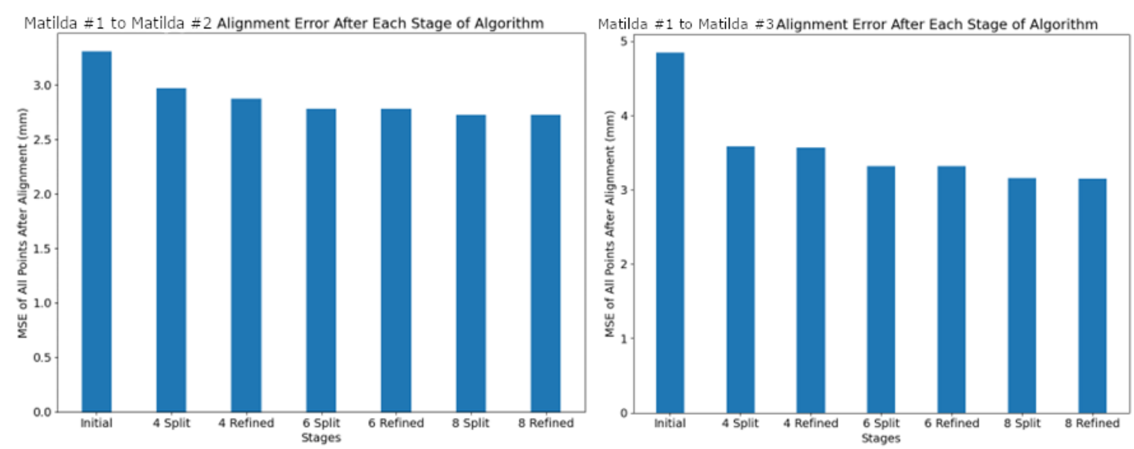


Figure 4.15: MSE at each stage of segmentation for various Matilda multiple pairs

The MSE data reinforces much of the previous analysis. Though the Matilda #1-Matilda #3 pair being with much more alignment error than the Matilda #1-Matilda #2 pair, segmentation yields a much greater reduction in MSE, indicating that the segments are more meaningful for

Matilda #1-Matilda #3. Like with the Horses, the re-evaluation step seems to have utility at the 4-segment phase but again doesn't successfully reduce MSE for larger amounts of segments. The MSE again shows the effects of diminishing returns, though again determining a threshold to use is a challenge, especially in this case where the "correct" stopping point is uncertain given the true number of segments is a bit uncertain.

4.3 St. Agnes Multiples

The St. Agnes multiples are also attributed to Bernini, with two scanned multiples in the dataset. The two multiples are St. Agnes #1 and St. Agnes #2, as seen in Fig. 4.16. No "true" segmentation scheme was considered for these multiples, so no accuracy analysis was performed.

Figure 4.16: St. Agnes multiples

4.3.1 Segmentations

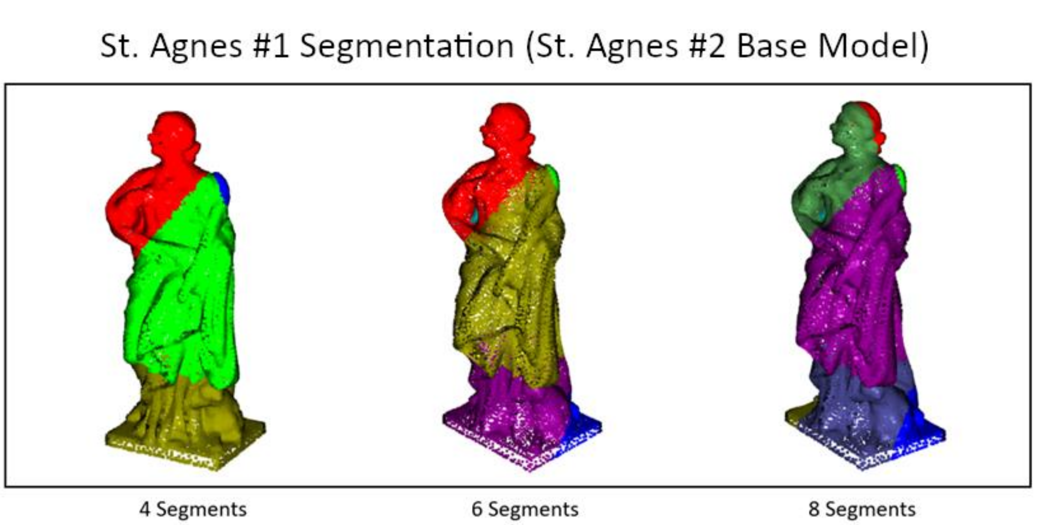


Figure 4.17: St. Agnes #1 to St. Agnes #2 segmentations

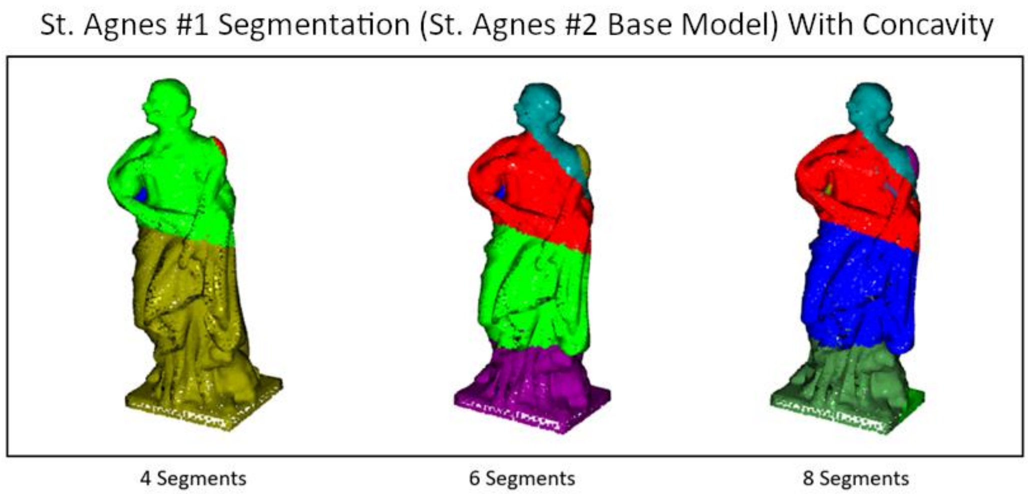


Figure 4.18: St. Agnes #1 to St. Agnes #2 segmentations factoring in concavity

Visually inspecting the segmentations in Fig. 4.17 and Fig. 4.18, some clear insights are available. There is a clear segment of the dress jutting out near the shoulder for only one of the models, so much so that the segmentation identifies and segments that small area immediately. The lower part of the dress and the lamb are segmented, with an additional segment on the upper body only when using concavity. No segmentation includes just the head, as with and without concavity the head is segmented with either the right or left arm, perhaps indicating the heads to not meaningfully deviate in a way the models can pick up.

Notably, the preprocessing step fills holes under the right and left arms that were missed in the scans, and these regions are identified and isolated, especially in the concavity segmentation where both are identified in the 8-segment version. This is a case where hole fixing can not fix the lack of data in those regions. Though the hole is "filled in" with faces, the points are still missing from the region, and when comparing with a multiple that is not missing these points there's great opportunity for misalignment, in the same way that the fully missing baton in the Matildas caused misalignment.

Overall, it's hard to say if any meaningful segments are derived in the images above.

4.3.2 Evaluations

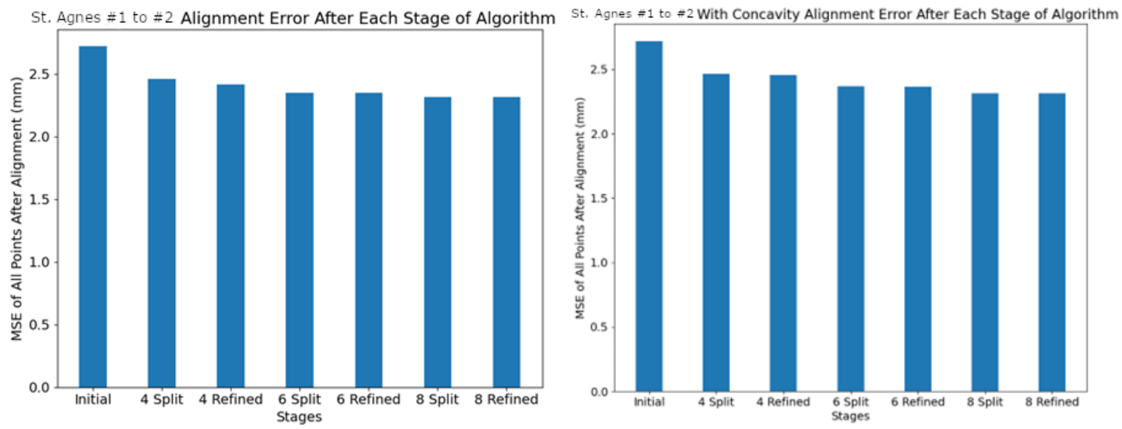


Figure 4.19: MSE at each stage of segmentation for various St. Agnes multiple pairs

The most notable aspect of the MSE data is the lack of reduction from the initial alignment to the 4-segment alignment. It's clear that even the first set of segmentations do not identify a clear misalignment of segments, and the diminishing returns of segmentation are immediately present. Of course, this does not confirm that both bronzes were casted in one piece, only indicating that detecting meaningful misalignments if there are joins present would require reducing the noise in the data (better scans without holes, size normalization).

4.4 Crucifix Multiples

The Crucifixes are a set of two multiples within the Bernini's Bronzes dataset, Crucifix #1 and Crucifix #2, as seen in Fig. 4.20. There are no known segments used for this analysis, though a fair amount is known about their casting process from the technical research.

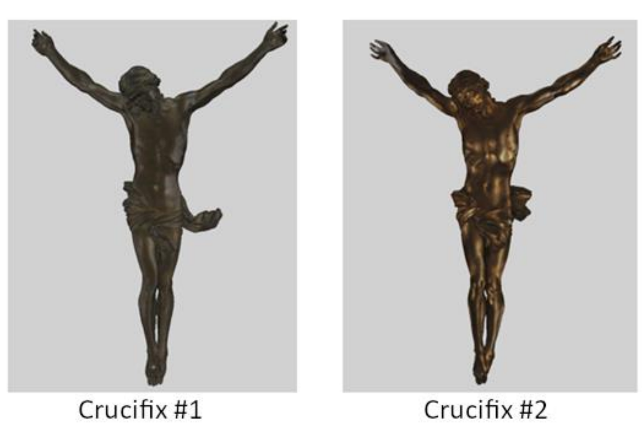


Figure 4.20: Crucifix multiples

4.4.1 Segmentations

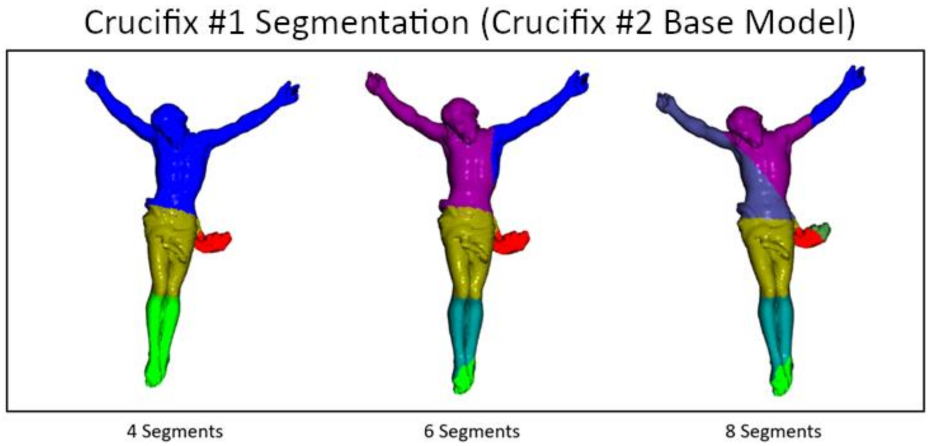


Figure 4.21: Crucifix #1 to Crucifix #2 segmentations

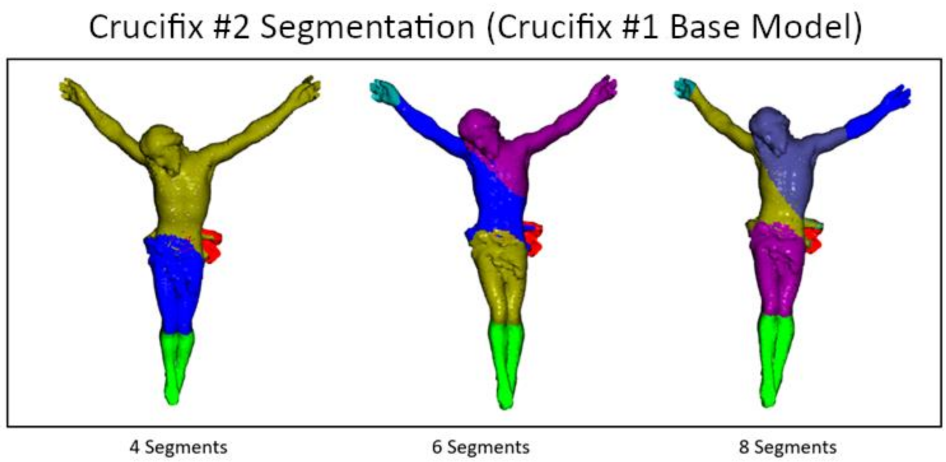


Figure 4.22: Crucifix #2 to Crucifix #1 segmentations

Two important regions to consider in the Crucifixes are the legs and the drapery. Evidence the legs and drapery of Crucifix #1 are entirely unique from the corresponding segments of Crucifix #2. The components do not originate from the exact same original mould by Bernini, and thus violate the assumptions held by the algorithm, that the segments themselves are nearly identical.

This means that all segmentation results within these components, like at the feet or the knees of the legs, are not meaningful for detecting joins. The segments are only useful for aligning these unalike components (one set of legs are more bent at the knees, for example). These regions generally hurt the performance of the algorithm as they require oversegmenting the disparate components to align them, which means the cost function will more often be minimized in those regions instead of the more meaningful segments of the rest of the model.

Additionally, the actual join between the legs and the torso is a sleeve, where the legs are slotted in within the drapery outer ring. The segmentation seam for the legs identified by the algorithm is above the drapery, as the drapery itself adds a lot of complexity in the region and prevents a planar segment from cleanly cutting at the top of the legs. Complex regions in general can cause issues with the algorithm as the limitation of planar seams is exposed.

As for the rest of the segments, the arms are clearly a point of interest, though interestingly the right arm is cut at the elbow in both pairwise segmentations. On the other hand, the left arm is grouped with the lower torso. Whether this can be attributed to noise as was likely in the St. Agnes multiples, or there is some other meaning in that selection of seam, it clearly does not identify a wax-to-wax join.

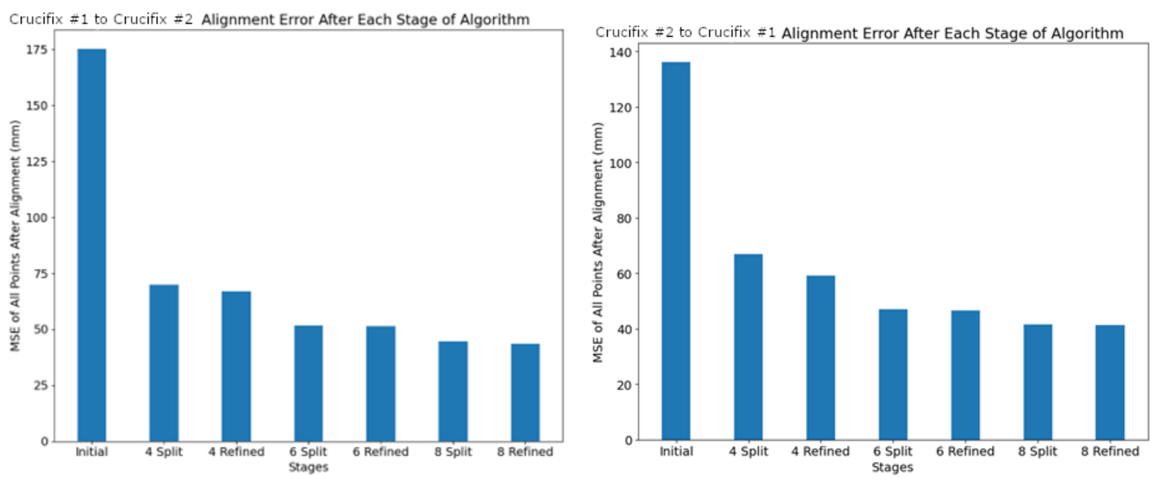


Figure 4.23: MSE at each stage of segmentation for various Crucifix multiple pairs

The initial MSE value of these models is easily the highest of any of the multiples studied in this paper, entirely due to the disparate legs and drapery. The first four segments clearly resolve much of the deviation, though the subsequent segmentations do not decay in MSE reduction nearly as much as some of the others, perhaps indicating that there's some meaning to the later segments.

For direction of alignment, the Crucifix #1-Crucifix #2 direction generally has higher alignment errors due to Crucifix #1's drapery sticking much further outline, but the trend of both as segmentation is performed does not indicate a meaningful asymmetry.

5. Conclusion and Next Steps

Overall, the results of this paper seem to indicate some utility in this algorithmic approach that can be leveraged by the Bernini's Bronzes technical study to learn more about the process and authorship of the bronze multiples.

For the clean, well-understood test dataset on the Horses, the algorithm shows a great ability to detect deviations between multiples and use those deviations to find the approximate locations of real segment joints. With ideal conditions, returning upwards of 90% accuracy on classifying points compared to the real segments is a great start, and where performance issues occur they still provide insights into the geometries of the models.

On the Bernini's Bronze data, performance seems to suffer due to the additional complexities and uncertainty around many aspects of the data, but still there is lots of interesting insights revealed. The Matildas in particular strongly indicate segments that match with hands-on expert analysis on the bronzes, helping to reinforce these observations. The results on the St. Agnes reveal interesting aspects of the multiples with their lack of results, hopefully driving additional discussion. The Crucifixes also provide nuanced points of exploration, specifically the arm segments selected and how those deviations tell the story of the multiples' casting.

Many potential improvements to both the algorithm and the methodology are evident from this work, most importantly applying size normalization to all compared multiples. Many of the segmentation results were considerably worse when a size difference was present, as oversegmenting is rewarded due to the larger gaps between the surfaces. Normalizing size would require some measure of volumetric comparison between the multiples, and may interact with other research from the team on shrinkage and how it manifests in bronzes.

Another major avenue for improvement involves the preprocessing stage, or even earlier in data collection. The solutions posed by this paper to holes and other scan artifacts allow the algorithm to run more easily, but holes in particular still cause artificial deviations between multiples. This was a large concern for the St. Agnes multiples in particular, where holes directly caused regions to be segmented. Either improving the hole selection process to better align the multiples or improving the scan data itself would greatly benefit further research in this area.

There is also great potential for improving the segmentation candidate generation. Many geometries encountered during this research, including the torus topology of some of the horse models and the complex drapery of the crucifixes, present major obstacles for the current system of generating seams, as planar cuts can not adequately segment the meshes in these regions. Even in non-problematic areas, true segments aren't often entirely planar, so if the goal is to approximate these as best as possible, more flexibility in drawing seams is necessary. Given just how large the possibility space of segmentation seams is, this is not an easy task, but any improvement, even another planar system that runs more quickly and arrives at the best segments fast, would provide value.

Finally, numerically defining a stopping criterion for segmentation is another priority. Visual analysis of the segmentation results and the RMSE graphs reveal intuitive stopping points, but this paper does not arrive at a definitive rule for when to stop segmenting on multiples with unknown segments. Given that this research is inherently exploratory and meant to support other technical analysis, this is not the highest priority issue to solve, but establishing some sort of confidence interval in each new segment would help with interpretation of results.

The hope for this research is that it supports the technical analysis of the Bernini's Bronzes team and drives this niche area of research further. Identifying regions of interest, approximating real segmentation joints, and drawing attention to odd relationships between multiples is the end goal, and ideally the art historical focus of the study is aided by this approach to analyzing alignment and deviation in multiples.

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